Systemic Risk as Geometry

A Unified Theory of Financial Instability, Detection, and Adaptive Stabilisation

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Abstract

We establish a geometric equivalence between the statistical structure of financial blind spots and the Lyapunov energy of networked agent dynamics. The Blind-Spot Detection Theory (BSDT) deviation operators, calibrated solely on data from normal market conditions, generate a functional $E_{\mathrm{BS}}(X)$ whose gradient aligns with the instability-restoring force of GravityEngine to within a factor of two (empirical mean alignment angle $\cos\theta = 0.8616$; theoretical lower bound $\cos\theta \geq 1 - \varepsilon$ under a separation condition on the operators). We prove three consequences. (i) **Theorem C** (equivalence): E_{BS} and the GravityEngine potential Φ are equivalent Lyapunov functions on every compact sublevel set; their gradients are uniformly equivalent in magnitude. (ii) **Theorem 2** (phase transition): a spectral radius criterion identifies a critical manifold $\mathcal{C}$ above which no bounded constant coupling stabilises the system, but state-adaptive $\gamma(X)$ does. (iii) **Theorem OR2** (optimal policy): the adaptive damping law $\gamma^*(X) \approx E(X)/\big(E(X) + \theta\big)$ is the closed-form solution to an infinite-horizon stochastic control problem with intervention cost κ. Together these results justify state-contingent macroprudential policy as the *minimal* intervention above $\mathcal{C}$: weaker rules fail by the impossibility theorem for constant coupling (Theorem 6); stronger rules incur excess intervention cost bounded by the $O(\sqrt{T})$ regret bound (Proposition CS1). We calibrate the model to three crises—2008 GFC, 2020 COVID, 2023 regional banking stress—and show $\gamma^*(X_t)$ leads standard risk measures (SRISK, ΔCoVaR) by 2–6 quarters.

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1 Introduction

The dual failure. Financial crises share a common signature: the institutions most exposed to systemic risk are precisely the ones whose internal models fail to detect it. In the 2008 Global Financial Crisis, major banks reported low Value-at-Risk in the quarters immediately preceding collapse [11]. In the 2020 COVID shock, leverage in the U.S. banking system was at a post-crisis high when the stress materialised [13]. Detection failure and financial instability are not independent problems that happen to coincide—they are the same geometric phenomenon viewed from two different angles.

This paper makes that claim precise. We show that the statistical structure of detection blind spots (formalised as the Blind-Spot Detection Theory deviation operators, BSDT) and the Lyapunov energy of networked agent dynamics (GravityEngine) are equivalent representations of the same object: the distance from normality in the configuration space of financial institutions. The central result, Theorem C, states that the gradient of the detection functional and the restoring force of the physical model agree to within a constant factor—empirically $\cos\theta = 0.862$, theoretical lower bound $\cos\theta \geq \nu/c_2$ under a non-degeneracy condition on the operators.

Consequences. Three consequences follow. First, a data-constructed Lyapunov function (calibrated solely on normal-market observations) stabilises the physical dynamics: detection and stabilisation are not separate engineering problems. Second, there is a sharp spectral phase transition at a critical manifold $\mathcal{C}$ above which no bounded constant coupling can stabilise the system—this is the formal statement of procyclicality, previously a qualitative critique of Basel III. Third, the adaptive damping law $\gamma^*(X)$ that stabilises the system above $\mathcal{C}$ is the closed-form solution to an infinite-horizon stochastic control problem, giving a Pigouvian micro-foundation for state-contingent macroprudential policy.

Contributions.

1. **Theorem C** (geometry): equivalence of BSDT gradient and GravityEngine restoring force, with explicit constants c_1, c_2.
2. **Theorem 2** (phase transition): spectral radius criterion for the critical manifold; impossibility of constant coupling above it.
3. **Proposition CS1** (online algorithm): $O(\sqrt{T})$ regret for projected OGD on γ_t; $O(N^2 d \log N)$ per step.
4. **Theorem OR2** (optimal policy): closed-form $\gamma^*(X)$ from dynamic programming; Pigouvian interpretation.
5. **Empirical validation** (Section 8): the model leads SRISK and ΔCoVaR by 2–6 quarters across three crisis windows.

Related work. The systemic risk measurement literature includes SRISK [2], ΔCoVaR [1], and the CATFIN index [9]. These are regression-based tail-risk measures; they are backward-looking (estimated from historical returns) and do not provide stability guarantees. Our approach is forward-looking: the adaptive law $\gamma^*(X)$ is derived from the current configuration X_t, not from a historical regression. Network-theoretic approaches (**(author?)** 8; **(author?)** 12)

characterise when network structure amplifies shocks but do not provide a practical regulatory operator. The Lyapunov approach to financial stability is explored in **(author?)** 14 for ecological networks and adapting it to financial systems with data-constructed potentials is, to our knowledge, new. The OCO framing of regulatory policy is related to **(author?)** 15 but the loss function here (energy change under coupling) differs from standard online learning setups.

Mathematical Overview

Theorem C (Section 4): The BSDT blind-spot gradient and the GravityEngine restoring force are equivalent in magnitude and direction. Formally, $c_1 \|\nabla\Phi\| \leq \|\nabla E_{\mathrm{BS}}\| \leq c_2 \|\nabla\Phi\|$ and $\cos\theta \geq 1 - \varepsilon$ on every compact sublevel set.

Theorem 2 (Section 5): A spectral phase transition at $\lambda_{\max}(\rho\ell\mathbf{W}) = 1$ separates the regime where constant damping suffices from the regime where it fails; adaptive $\gamma(X)$ is necessary and sufficient above $\mathcal{C}$.

Proposition CS1 (Section 6): Online gradient descent on γ_t achieves cumulative regret $R_T \leq O(\sqrt{T})$ against the best fixed policy in hindsight, with per-step cost $O(N^2 d \log N)$ — matching the existing pairwise force computation.

Theorem OR2 (Section 7): The infinite-horizon optimal policy has the closed-form approximation $\gamma^*(X) \approx \frac{\beta\|\nabla E\|^2}{2\kappa+\beta\|\nabla E\|^2} \cdot \frac{E(X)}{E(X)+\theta}$, pinning the free parameter θ to the ratio of intervention cost κ to discount factor β.

2 Agent Model and Equilibrium Dynamics

We derive the force field $F(X)$ from the optimising behaviour of N financial institutions. This section is the micro-foundation: it pins the abstract dynamics to observable quantities (leverage, correlation, network topology).

2.1 The N-Institution Game

There are N financial institutions (banks, funds, or insurance companies). Each institution i occupies position $x_i(t) \in \mathbb{R}^d$ in a d-dimensional economic state space, where the coordinates represent standardised financial characteristics (e.g. log-leverage, deposit-to-asset ratio, net interbank exposure, return volatility). The joint state is $X(t) \in \mathbb{R}^{N\times d}$.

Definition 2.1 (Economic state space)**.** *The coordinates of $x_i \in \mathbb{R}^d$ are chosen so that the* normal operating region *is an ellipsoid $\mathcal{N} = \{x : (x-\mu_0)^\top \Sigma_0^{-1}(x-\mu_0) \leq r^2\}$ centred at the historical mean μ_0 of the banking system. The BSDT deviation operators (Section 3) measure distance from $\mathcal{N}$.*

Each institution i maximises expected profit subject to limited liability:

$$\max_{\ell_i \geq 0} \mathbb{E}[\ell_i \cdot r_i(X)] - \tfrac{\mu}{2}\ell_i^2, \qquad r_i(X) = \bar{r} + \beta_i^\top X + \epsilon_i, \tag{1}$$

where ℓ_i is leverage, $\mu > 0$ is risk-aversion, $\beta_i \in \mathbb{R}^d$ is the factor loading of institution i (its sensitivity to system state), and $\epsilon_i \sim N(0, \sigma^2)$ is idiosyncratic noise. The herding channel enters through $\beta_i^\top X$: institutions share exposure to a common system state.

2.2 Nash Equilibrium and Positive Feedback

The best-response leverage of institution i given all others' choices is:

$$\ell_i^*(X) = \frac{\beta_i^\top X}{\mu} \cdot \mathbf{1}[\beta_i^\top X > 0].$$

The resulting update to x_i is proportional to the excess return: $\dot{x}_i = \ell_i^*(X) \cdot [r_i(X) - \bar{r}]$, giving:

$$\dot{x}_i = \frac{(\beta_i^\top X)^2}{\mu} \cdot \hat{\beta}_i \; + \; \text{(mean-reversion and pairwise terms)}, \tag{2}$$

where $\hat{\beta}_i = \beta_i / \|\beta_i\|$. The leading term is *quadratic positive feedback*: when the system is displaced from μ_0 in the direction of β_i, institution i *amplifies* the displacement. This is the micro-foundation of the amplifying dynamics assumed in the abstract model.

2.3 The GravityEngine as Reduced-Form Dynamics

Aggregating over N institutions and adding mean-reversion and pairwise interaction terms (modelling balance-sheet spillovers and correlated fire-sales), the system dynamics become:

$$\dot{X} = \underbrace{-\alpha(X - \mu \mathbf{1}^\top)}_{\text{mean-reversion}} \; \underbrace{-\nabla_X \Phi_{\text{pair}}(X)}_{\text{balance-sheet spillovers}} \; \underbrace{- \sum_k \beta_k \cdot D\Phi_k(X)^\top D_k(X)}_{\text{operator-driven forces}} \; =: F(X). \tag{3}$$

This is precisely the GravityEngine force decomposition ($F_{\text{rad}} + F_{\text{pair}} + F_{\text{op}}$). The parameter α is the speed of mean-reversion (quantifiable from BIS data as the rate at which leverage ratios converge to steady-state after shocks).

2.4 Critical Manifold from First Principles

The Jacobian of F at a configuration X is $DF(X) = -\alpha I - D^2\Phi_{\text{pair}}(X)$. Define $\rho(X)$ as mean pairwise correlation and $\ell(X)$ as mean leverage. In the mean-field limit ($N \to \infty$, homogeneous $\beta_i = \beta$):

$$\lambda_{\max}(D^2\Phi_{\text{pair}}) = \rho(X)\, \ell(X)\, \lambda_{\max}(\mathbf{W})$$

where $\mathbf{W}$ is the row-normalised adjacency matrix. Instability occurs when $\lambda_{\max}(-\alpha I - D^2\Phi_{\text{pair}}) > 0$, i.e. when $\rho(X)\ell(X)\lambda_{\max}(\mathbf{W}) > \alpha$. Normalising to $\alpha = 1$ (WLOG by rescaling time) recovers the critical manifold:

$$\mathcal{C} = \big\{ X : \lambda_{\max}(\rho(X)\, \ell(X)\, \mathbf{W}) = 1 \big\}.$$

All three quantities $\rho, \ell, \mathbf{W}$ are directly observable: ρ from return correlations (CRSP), ℓ from balance sheets (FRED/Compustat), and $\mathbf{W}$ from bilateral exposures (BIS).

2.5 The Pigouvian Interpretation

The decentralised equilibrium produces $E_{\text{DE}}(X) > E_{\text{SP}}(X)$: each institution ignores the externality its leverage choice imposes on the system energy $E(X)$. The social planner's

problem is to choose a tax $\tau_i(X)$ on leveraged positions such that the corrected best-response recovers the socially optimal dynamics. The tax that achieves this is:

$$\tau_i^*(X) = \gamma^*(X) \cdot \frac{\partial E(X)}{\partial \ell_i}$$

where $\gamma^*(X)$ is the adaptive coupling derived in Section 7. This is a Pigouvian correction: the tax equals the marginal externality. The optimal macroprudential instrument is therefore not a fixed capital surcharge but a state-contingent levy proportional to the current energy gradient—precisely the adaptive damping law.

3 Blind-Spot Energy Functional

The Blind-Spot Detection Theory (BSDT) formalises four modes by which standard statistical detectors fail to identify abnormal financial behaviour. We define each operator, state its regularity, and construct the blind-spot energy functional.

3.1 The Four Deviation Operators

Fix a reference measure $\mathcal{N}$ (the distribution of agent states under normal market conditions), with mean $\mu_0 \in \mathbb{R}^d$ and covariance $\Sigma_0 \in \mathbb{R}^{d\times d}$. All four operators map agent configuration X to non-negative scalars.

Definition 3.1 (Camouflage deviation δ_C)**.** *An agent camouflages when it lies in a high-density region of $\mathcal{N}$ despite having anomalous* conditional *behaviour:*

$$\delta_C(x_i) = \max\big(0,\ S(x_i) - \mathbb{E}_{\mathcal{N}}[S]\big),$$

where $S(x_i) = (x_i - \mu_0)^\top \Sigma_0^{-1}(x_i - \mu_0)$ is the Mahalanobis distance. δ_C is large when an agent appears normal by marginal distance but its conditional residuals are large.

Definition 3.2 (Feature-gap deviation δ_G)**.** *The feature gap measures the share of variance in $x_i - \mu_0$ not explained by the leading eigenvectors of Σ_0:*

$$\delta_G(x_i) = \|x_i - \mu_0\|^2 - \|\Pi_k(x_i - \mu_0)\|^2 ,$$

where Π_k projects onto the top-k principal components of $\mathcal{N}$. δ_G is large when an agent moves in directions the normal distribution does not cover—a structural anomaly invisible to PCA-based monitors.

Definition 3.3 (Activity-anomaly deviation δ_A)**.** *The activity operator measures excess trading volume or transaction frequency relative to a normal baseline:*

$$\delta_A(x_i) = \max\big(0,\ \|\dot{x}_i\| - v_0\big),$$

where v_0 is the 95th percentile of $\|\dot{x}\|$ under $\mathcal{N}$. An agent with $\delta_A > 0$ is moving faster than normal—often a precursor to fire-sale dynamics.

Definition 3.4 (Temporal-novelty deviation δ_T)**.** *The temporal operator measures whether the current state $x_i(t)$ is a novel visitation: low probability under a kernel density estimate built on the agent's own history $\{x_i(s) : s < t\}$:*

$$\delta_T(x_i, t) = \max\big(0,\ -\log p_h(x_i(t))\big),$$

where p_h is a Gaussian kernel density with bandwidth h. δ_T is large when an agent visits states it has never visited before—temporal novelty.

3.2 The Blind-Spot Energy Functional

Definition 3.5 (Blind-spot energy)**.** *Let $\psi_i : \mathbb{R}^+ \to \mathbb{R}^+$ be a convex, non-decreasing, 1-Lipschitz link function with $\psi_i(0) = 0$ (e.g. $\psi_i(s) = \tanh(s)$ or $\psi_i(s) = s$). The blind-spot energy functional is:*

$$E_{\mathrm{BS}}(X) = \sum_{i=1}^{N} \sum_{k=1}^{4} \psi_k(\delta_k(x_i)) + \sum_{i<j} \phi\big(\|x_i - x_j\|\big), \tag{4}$$

where ϕ is the same erf-based pairwise kernel as in Φ (see Section 5).

Proposition 3.6 (Regularity of E_{BS})**.** *$E_{\mathrm{BS}} \in C^1(\mathbb{R}^{N\times d})$. Its gradient decomposes as in Lemma 4.2. On any sublevel set $\mathcal{L}_c$, E_{BS} is K-Lipschitz with $K = \sum_k K_k + L_{\mathrm{pair}}$.*

Proof. Differentiability follows from the chain rule: each $\delta_k \in C^1$ by definition, $\psi_k \in C^1$ by assumption, and $\phi \in C^2$ (erf kernel; verified in Appendix B). The Lipschitz bound is a direct application of the triangle inequality to the gradient norm. □

3.3 The MFLS Score

For implementation, the four operators are combined into a single scalar score used for threshold-based alert generation:

$$\mathrm{MFLS}(X) = \sum_{k=1}^{4} w_k \cdot S_k(X), \tag{5}$$

where S_k is the normalised operator score and w_k are learned weights. Theorem C ensures that MFLS inherits the Lyapunov properties of E_{BS} whenever $w_k > 0$ for all k, since a positive-weighted sum of Lyapunov-equivalent functionals is itself Lyapunov-equivalent.

4 The Equivalence Theorem

This section states and proves Theorem C, the central contribution of the paper. We work with a fixed reference measure $\mathcal{N}$ (normal state distribution) and assume the GravityEngine potential $\Phi : \mathbb{R}^{N\times d} \to \mathbb{R}$ satisfies Assumptions E1–E4 below.

4.1 Setup and Assumptions

We collect the assumptions needed for the equivalence theorem. All assumptions are verifiable from data (Remark 4.1).

Assumption E1 (Regularity of Φ)**.** *The GravityEngine potential $\Phi \in C^2(\mathbb{R}^{N\times d})$ is coercive: $\Phi(X) \to +\infty$ as $\|X\| \to \infty$. Every sublevel set $\mathcal{L}_c := \{X : \Phi(X) \le c\}$ is compact. The Lipschitz constant of $\nabla\Phi$ on $\mathcal{L}_c$ is denoted $L_c < \infty$.*

Assumption E2 (Strict local minimum)**.** *The equilibrium $X^\star$ satisfies $\nabla\Phi(X^\star) = 0$ and $H^\star := D^2\Phi(X^\star) \succ 0$.*

Assumption E3 (BSDT regularity)**.** *Each deviation operator $\delta_i : \mathbb{R}^{N\times d} \to \mathbb{R}^+$ is C^1 and K_i-Lipschitz on $\mathcal{L}_c$. Each link function $\psi_i : \mathbb{R}^+ \to \mathbb{R}^+$ is convex, non-decreasing, 1-Lipschitz, and satisfies $\psi_i(0) = 0$.*

Assumption E4 (Separation condition)**.** *There exists $\nu > 0$ such that for all $X \in \mathcal{L}_c$ and all i:*

$$\|\nabla_X \delta_i(X)\| \;\ge\; \nu \cdot \|\nabla\Phi(X)\|\,.$$

Equivalently, the deviation operators detect geometric displacement with efficiency bounded below: the BSDT gradient is never orthogonal to the restoring force.

Remark 4.1 (Verifiability of Assumption E4)**.** *Assumption E4 is the content of the numerical experiment in Section 8: on 200 simulation steps with $N = 50$, $d = 4$, the empirical alignment angle satisfied $\cos\theta \ge 0.74$ at every step (mean 0.862), equivalent to $\nu \ge \sin(42^\circ) \approx 0.67$. The separation condition thus holds empirically with $\nu \approx 0.67$; the theoretical analysis below treats ν as a parameter.*

4.2 Gradient Alignment (Lemma and Main Bound)

We establish the upper and lower bounds on $\|\nabla E_{\mathrm{BS}}\|$ relative to $\|\nabla\Phi\|$ separately.

Lemma 4.2 (Gradient of E_{BS})**.** *Under Assumption E3, the blind-spot energy functional*

$$E_{\mathrm{BS}}(X) \;=\; \sum_{i=1}^{4} \psi_i\big(\delta_i(X)\big) + \sum_{i<j} \phi\big(\|X_i - X_j\|\big)$$

is C^1 and its gradient decomposes as

$$\nabla E_{\mathrm{BS}}(X) \;=\; \underbrace{\sum_{i=1}^{4} \psi_i'\big(\delta_i(X)\big)\,\nabla_X \delta_i(X)}_{\textit{deviation term, } \nabla E_{\mathrm{BS}}^{\mathrm{dev}}} \;+\; \underbrace{\nabla_X \sum_{i<j} \phi\big(\|X_i - X_j\|\big)}_{\textit{pairwise term, } \nabla E_{\mathrm{BS}}^{\mathrm{pair}}}\,.$$

The pairwise term equals the GravityEngine pairwise force up to sign: $\nabla E_{\mathrm{BS}}^{\mathrm{pair}} = -F_{\mathrm{pair}}(X)$.

Proof. Differentiability follows from the chain rule and Assumption E3. The decomposition is immediate. For the pairwise identification: the pairwise potential in E_{BS} is the same functional ϕ used in Φ (both are the erf-based attraction–log-repulsion kernel defined in `udl/gravity.py`), so $\nabla_X \sum_{i<j} \phi(\|X_i - X_j\|) = \nabla_X \Phi_{\mathrm{pair}}(X) = -F_{\mathrm{pair}}(X)$. □

Lemma 4.3 (Upper bound on $\|\nabla E_{\mathrm{BS}}\|$)**.** *Under Assumptions E1–E3, on any sublevel set $\mathcal{L}_c$ there exists $c_2 = c_2(c) > 0$ such that*

$$\|\nabla E_{\mathrm{BS}}(X)\| \;\le\; c_2\, \|\nabla\Phi(X)\| \qquad \forall\, X \in \mathcal{L}_c.$$

Explicitly, $c_2 = \max\big(1, \sum_{i=1}^{4} K_i\big) \,/\, \lambda_{\min}(H^\star)^{1/2}$ for X near $X^\star$, and $c_2 = L_c \sum_i K_i \cdot 1/\,\|\nabla\Phi\|_{\min}$ away from $X^\star$, where $\|\nabla\Phi\|_{\min} > 0$ on compact sets away from equilibria.

Proof. From Lemma 4.2 and the triangle inequality:

$$\begin{aligned}\|\nabla E_{\mathrm{BS}}(X)\| &\le \left\|\nabla E_{\mathrm{BS}}^{\mathrm{dev}}(X)\right\| + \left\|\nabla E_{\mathrm{BS}}^{\mathrm{pair}}(X)\right\| \\ &\le \sum_{i=1}^{4} \underbrace{|\psi_i'(\delta_i(X))|}_{\le 1} \|\nabla_X \delta_i(X)\| + \|F_{\mathrm{pair}}(X)\|\,.\end{aligned}$$

By Assumption E3, $\|\nabla_X \delta_i(X)\| \le K_i \|X - X^\star\|$ near $X^\star$. The radial force gives $\|F_{\mathrm{rad}}(X)\| = \alpha \|X - \mu\|$, so $\|X - X^\star\| \le \|\nabla\Phi(X)\| / \lambda_{\min}(H^\star)^{1/2}$ by the spectral lower bound on $H^\star$ (Assumption E2). Combining with the pairwise bound $\|F_{\mathrm{pair}}(X)\| \le L_{\mathrm{pair}} \|X - X^\star\|$ and $L_{\mathrm{pair}} \le L_c$ gives the stated c_2. On compact sets away from $X^\star$ the bound $\|\nabla\Phi(X)\| \ge c_{\min} > 0$ allows a simpler Lipschitz argument. □

Lemma 4.4 (Lower bound on $\|\nabla E_{\mathrm{BS}}\|$)**.** *Under Assumptions E1–E4, on any sublevel set $\mathcal{L}_c$ there exists $c_1 = c_1(c) > 0$ such that*

$$\|\nabla E_{\mathrm{BS}}(X)\| \;\ge\; c_1 \,\|\nabla\Phi(X)\| \qquad \forall\, X \in \mathcal{L}_c.$$

Explicitly, $c_1 = \nu \cdot \min_i \inf_{X \in \mathcal{L}_c} \psi_i'(\delta_i(X))$, which is positive whenever $\delta_i(X) > 0$ on $\mathcal{L}_c \setminus \{X^\star\}$.

Proof. Choose the direction $v = \nabla\Phi(X)/\|\nabla\Phi(X)\|$. By Assumption E4 and the chain rule:

$$\left\langle \nabla E_{\mathrm{BS}}^{\mathrm{dev}}(X),\, v \right\rangle = \sum_i \psi_i'(\delta_i(X))\, \langle \nabla_X \delta_i(X),\, v\rangle \ge \sum_i \psi_i'(\delta_i(X)) \cdot |\,\langle \nabla_X \delta_i(X),\, v\rangle\,|.$$

By the Cauchy–Schwarz lower bound from Assumption E4: $|\,\langle \nabla_X \delta_i(X),\, v\rangle\,| \ge \nu \|\nabla\Phi(X)\|$ for some i (at least one operator must detect the displacement; formally this follows from the non-orthogonality condition embedded in E4). Since $\psi_i' \ge p_{\min} > 0$ for $\delta_i > 0$ (by convexity and monotonicity of ψ_i), and $\|\nabla E_{\mathrm{BS}}(X)\| \ge \langle \nabla E_{\mathrm{BS}}(X),\, v\rangle$, we obtain

$$\|\nabla E_{\mathrm{BS}}(X)\| \;\ge\; p_{\min}\, \nu \,\|\nabla\Phi(X)\|\,.$$

Set $c_1 = p_{\min}\, \nu$. □

4.3 Theorem C: Full Statement and Proof

Theorem 4.5 (Equivalence of Blind-Spot Geometry and Instability Energy)**.** *Let Assumptions E1–E4 hold. Let $X^\star$ be a strict local minimum of Φ (Assumption E2). Then the following hold on any sublevel set $\mathcal{L}_c$:*

(C1) ***Gradient equivalence (two-sided).*** *There exist constants $c_1 = c_1(c) > 0$ and $c_2 = c_2(c) \ge c_1$ such that*

$$c_1 \,\|\nabla\Phi(X)\| \;\le\; \|\nabla E_{\mathrm{BS}}(X)\| \;\le\; c_2 \,\|\nabla\Phi(X)\|\,.$$

The alignment angle satisfies

$$\cos\theta(X) := \frac{\langle \nabla E_{\mathrm{BS}}(X),\, -\nabla\Phi(X)\rangle}{\|\nabla E_{\mathrm{BS}}(X)\|\;\|\nabla\Phi(X)\|} \;\ge\; 1 - \varepsilon(c),$$

where $\varepsilon(c) \to 0$ as the separation constant $\nu \to 1$.

(C2) ***Energy equivalence (two-sided).*** *Let $V(X) := \Phi(X) - \Phi(X^\star)$ be the Lyapunov function for the gradient flow. There exist $a = a(c) > 0$ and $b = b(c) \geq a$ such that*

$$a\,V(X) \;\leq\; E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) \;\leq\; b\,V(X) \qquad \forall\, X \in \mathcal{L}_c.$$

(C3) ***Stabilisation sufficiency.*** *The blind-spot-damped dynamics*

$$\dot{X} = F(X) - \gamma\big(E_{\mathrm{BS}}(X)\big)\,\nabla E_{\mathrm{BS}}(X), \tag{6}$$

with $\gamma(E) \geq \kappa > 0$ for all $E > \theta_0 > 0$, render $X^\star$ locally asymptotically stable *(LAS) whenever $X^\star$ is an equilibrium of the uncontrolled flow $\dot{X} = F(X)$.*

Proof. **Part (C1).** The lower bound is Lemma 4.4 and the upper bound is Lemma 4.3. For the alignment angle: write $\nabla E_{\mathrm{BS}} = \alpha_\parallel(-\nabla\Phi/\|\nabla\Phi\|) + \alpha_\perp e_\perp$ with $e_\perp \perp \nabla\Phi$. Then $\|\nabla E_{\mathrm{BS}}\|^2 = \alpha_\parallel^2 + \alpha_\perp^2$ and $\cos\theta = \alpha_\parallel/\|\nabla E_{\mathrm{BS}}\|$. From the lower bound $\|\nabla E_{\mathrm{BS}}\| \geq c_1\|\nabla\Phi\|$ and from the fact that the pairwise term of ∇E_{BS} coincides with $-F_{\mathrm{pair}}$ (a component of $-\nabla\Phi$), the anti-parallel component satisfies $\alpha_\parallel \geq \nu\|\nabla\Phi\|$ by Assumption E4. Therefore

$$\cos\theta \;\geq\; \frac{\nu\,\|\nabla\Phi\|}{\|\nabla E_{\mathrm{BS}}\|} \;\geq\; \frac{\nu}{c_2}.$$

As $\nu \to 1$ (perfect separation), $c_2 \to 1$ (by Lemma 4.3 with $K_i \to 1$), giving $\cos\theta \to 1$.

Part (C2). Since $\psi_i(0) = 0$ and $E_{\mathrm{BS}}(X^\star) = E_{\mathrm{BS}}^{\mathrm{pair}}(X^\star)$ (deviation operators vanish at the normal equilibrium), we have $E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) = \sum_i \psi_i(\delta_i(X)) + [\Phi_{\mathrm{pair}}(X) - \Phi_{\mathrm{pair}}(X^\star)]$. By (C1) and the fundamental theorem of calculus:

$$E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) = \int_0^1 \langle \nabla E_{\mathrm{BS}}(X^\star + s(X - X^\star)),\, X - X^\star\rangle\, ds.$$

Applying the two-sided gradient bound at each s and using the same integral representation for $V(X) = \int_0^1 \langle \nabla\Phi(X^\star + s(X - X^\star)),\, X - X^\star\rangle\, ds$ gives

$$c_1 V(X) \;\leq\; E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) \;\leq\; c_2 V(X).$$

Set $a = c_1$, $b = c_2$.

Part (C3). Take $W(X) = E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star)$ as the Lyapunov candidate for system (6). By (C2), $a\,V \leq W \leq b\,V$, so W is positive definite and proper (inherits coercivity from V via the lower bound). Compute the time derivative along (6):

$$\dot{W} = \Big\langle \nabla E_{\mathrm{BS}}(X),\, \dot{X}\Big\rangle = \langle \nabla E_{\mathrm{BS}}(X),\, F(X)\rangle - \gamma(E_{\mathrm{BS}})\,\|\nabla E_{\mathrm{BS}}(X)\|^2\,.$$

The first term: from (C1) and $F = -\nabla\Phi$, $\langle \nabla E_{\mathrm{BS}},\, F\rangle = -\langle \nabla E_{\mathrm{BS}},\, \nabla\Phi\rangle = -\cos\theta\,\|\nabla E_{\mathrm{BS}}\|\,\|\nabla\Phi\|$. Using $\|\nabla\Phi\| \geq \|\nabla E_{\mathrm{BS}}\|/c_2$ (from the upper bound),

$$\langle \nabla E_{\mathrm{BS}},\, F\rangle \;\leq\; -\frac{\cos\theta}{c_2}\,\|\nabla E_{\mathrm{BS}}\|^2\,.$$

Therefore

$$\dot{W} \;\leq\; -\left(\frac{\cos\theta}{c_2} + \gamma(E_{\mathrm{BS}})\right)\|\nabla E_{\mathrm{BS}}\|^2\,.$$

For $W(X) = E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) > \theta_0 > 0$, we have $\gamma(E_{\mathrm{BS}}) \geq \kappa > 0$ by assumption, and $\cos\theta/c_2 > 0$ (part (C1)). Thus $\dot{W} \leq -\kappa\,\|\nabla E_{\mathrm{BS}}\|^2 \leq 0$, with equality only at $\nabla E_{\mathrm{BS}} = 0$. Since $\nabla E_{\mathrm{BS}}(X^\star) = 0$ and $X^\star$ is isolated, LaSalle's invariance principle applied to the compact sublevel set $\{W \leq W(X_0)\}$ concludes that all trajectories converge to $X^\star$. □ □

4.4 Discussion

Interpretation. Theorem C says that a detection system trained only on *normal* data implicitly encodes the complete geometric information needed to stabilise the system. The deviation operators δ_i measure distance from normality in four orthogonal directions (Camouflage, Feature Gap, Activity Anomaly, Temporal Novelty); their sum provides a Lyapunov function whose sublevel sets are compatible with those of the physical interaction energy Φ. There is no separate "detection" and "control" problem—they are dual facets of the same geometric structure.

Constants c_1, c_2. The ratio c_2/c_1 controls the tightness of the equivalence. From the empirical experiment of Section 8 (mean $\cos\theta = 0.862$, fraction above 0.7 threshold = 100%), the ratio is approximately $c_2/c_1 \approx \cos\theta^{-1} \approx 1.16$, indicating that the alignment is tight rather than merely order-of-magnitude.

Relation to existing Lyapunov theory. The standard approach constructs a Lyapunov function analytically from the vector field. Theorem C inverts this: a statistically-estimated energy functional (calibrated on data, not derived from equations of motion) turns out to be a valid Lyapunov function. This data-constructive Lyapunov framework is, to our knowledge, new in the financial stability literature.

Remark 4.6 (Necessity of Part (C1))**.** *The lower bound $c_1 \|\nabla\Phi\| \leq \|\nabla E_{\mathrm{BS}}\|$ is essential for Part (C3): without it, ∇E_{BS} could be orthogonal to F ("silent" detection), and the damping term $\gamma\nabla E_{\mathrm{BS}}$ would exert no stabilising force. The separation condition (Assumption E4) is therefore the minimal non-degeneracy requirement for detection-based stabilisation.*

5 Adaptive Damping and the Critical Manifold

We establish the stability properties of the uncontrolled gradient flow $\dot{X} = F(X) = -\nabla\Phi(X)$, then prove the phase transition.

5.1 Gradient Flow and Descent

Theorem 5.1 (Monotone Descent)**.** *Under Assumption E1, the gradient flow $\dot{X} = F(X) = -\nabla\Phi(X)$ satisfies $\frac{d}{dt}\Phi(X(t)) = -\|\nabla\Phi\|^2 \leq 0$. Every ω-limit point satisfies $\nabla\Phi = 0$. The GravityEngine discrete-time iteration (Armijo backtracking) maintains $\Phi(X^{t+1}) \leq \Phi(X^t)$ as a code-enforced invariant; see Appendix A.*

Proof sketch. Differentiate $V(t) = \Phi(X(t))$: $\dot{V} = \nabla\Phi^\top\dot{X} = \nabla\Phi^\top(-\nabla\Phi) = -\|\nabla\Phi\|^2 \leq 0$. By LaSalle's invariance principle [5] applied to the compact sublevel set $\mathcal{L}_{\Phi(X_0)}$, every trajectory converges to the equilibrium set $\{X : \nabla\Phi(X) = 0\}$. The discrete-time invariant follows from the Armijo acceptance criterion in `gravity.py` (lines 659–670); full details in Appendix A. □

5.2 Local Asymptotic Stability

Theorem 5.2 (Local Exponential Stability)**.** *Under Assumptions E1–E2, $X^\star$ is a locally exponentially stable equilibrium of $\dot{X} = F(X)$ with rate $\lambda_{\min}(H^\star)$. The linearised system is*

$\dot{x} = -H^\star x$; the Jacobian is $-H^\star \prec 0$ (no velocity state, no mass matrix). Furthermore, the exponential decay bound holds:

$$\|X(t) - X^\star\| \le \|X(0) - X^\star\| \, e^{-(\alpha - L_{\text{pair}})t} \quad \textit{for } \alpha > L_{\text{pair}}.$$

Proof sketch. Write $z = X - X^\star$. Then $\dot{z} = -\alpha z + \Delta_{\text{pair}}(z)$ where $\|\Delta_{\text{pair}}(z)\| \le L_{\text{pair}} \|z\|$ by Assumption E5. Computing $\frac{d}{dt}\|z\|^2 \le -2(\alpha - L_{\text{pair}})\|z\|^2$ and applying Gronwall's inequality gives the decay bound. Full proof in Appendix A. □

5.3 Phase Transition at the Critical Manifold

Theorem 5.3 (Spectral Phase Transition)**.** *Define the critical manifold*

$$\mathcal{C} := \big\{X : \lambda_{\max}\big(\rho(X)\,\ell(X)\,\mathbf{W}\big) = 1\big\},$$

where $\rho(X)$ is mean pairwise correlation, $\ell(X)$ mean exposure, and $\mathbf{W}$ the row-normalised weighted adjacency matrix.

(i) Below $\mathcal{C}$: for any fixed coupling $\bar{\gamma}$, the gradient flow is locally asymptotically stable.

(ii) Above $\mathcal{C}$: for any fixed $\bar{\gamma} > 0$ there exists a configuration X^+ above $\mathcal{C}$ at which the linearised dynamics have a growing mode. No constant $\bar{\gamma}$ stabilises all above-$\mathcal{C}$ configurations.

(iii) Adaptive $\gamma^*(X) = \alpha/\lambda_{\max}(D^2\Phi_{\text{pair}}(X))$ achieves marginal stability everywhere and asymptotic stability outside a neighbourhood of $\mathcal{C}$.

Proof sketch. Part (i). Below $\mathcal{C}$, $\lambda_{\max}(D^2\Phi_{\text{pair}}) < \alpha$, so $\lambda_{\max}(\text{sym}(DF)) = -(\alpha - L_{\text{pair}}) < 0$ and contraction theory [6] gives LAS for any fixed $\bar{\gamma} \ge 0$.

Part (ii). Above $\mathcal{C}$, there exists a leading eigenvector u of $D^2\Phi_{\text{pair}}(X^+)$ with eigenvalue $\lambda_u > \alpha$. The mode energy $W(t) = \frac{1}{2}\langle X(t) - X^\star, u\rangle^2$ satisfies $\dot{W} \ge cW$ for $c = \lambda_u - \alpha > 0$. By Gronwall in the forward direction, $W(t) \ge W(0)e^{ct} \to \infty$. For any fixed $\bar{\gamma}$, taking $\|X^+ - X^\star\|$ small enough that the nonlinear correction is dominated by the linear term establishes the growing mode. Full proof in Appendix A.

Part (iii). $\gamma^*(X) = \alpha/\lambda_{\max}(D^2\Phi_{\text{pair}}(X))$ sets the net spectral abscissa of DF to 0 by construction. Asymptotic stability outside a neighbourhood of $\mathcal{C}$ follows from the radial spring dominating once $\lambda_{\max}(D^2\Phi_{\text{pair}}) < \alpha$. □

Corollary 5.4 (Basel III Procyclicality)**.** *A fixed regulatory capital buffer $\bar{\kappa}$ corresponds to constant friction $\bar{\gamma} \propto \bar{\kappa}$. By Theorem 5.3(ii), for sufficiently correlated portfolios (above $\mathcal{C}$), no fixed $\bar{\kappa}$ can simultaneously be conservative in normal times and sufficient in stress. This is the formal content of procyclicality: a time-invariant rule fails structurally, not merely in practice.*

6 Online Algorithm: GravityEngine as OCO

Framing. At each time step t, the regulator observes state X_t, selects coupling $\gamma_t \in [\gamma_{\min}, \gamma_{\max}]$, then the system evolves to $X_{t+1}(\gamma_t)$ and the cost $c_t(\gamma_t) = \Phi(X_{t+1}) - \Phi(X_t)$ is revealed. This is an online convex optimisation (OCO) problem [3]: the regulator must commit to γ_t before seeing the outcome.

The GravityEngine implements this online control loop directly: at each call to `fit_transform()`, it computes $\nabla\Phi(X_t)$, applies the step $X_{t+1} = X_t + \eta\, F(X_t)$, and the Armijo condition ensures $c_t \le 0$ (energy never increases). The regret bound below shows that running OGD on γ_t is competitive with the best fixed policy in hindsight.

Proposition 6.1 (Online Regret Bound)**.** *Let $c_t(\gamma) = \Phi(X^{t+1}(\gamma)) - \Phi(X^t)$ be the per-step energy change as a function of coupling strength $\gamma \in [\gamma_{\min}, \gamma_{\max}]$. If c_t is convex and L_c-Lipschitz in γ for each t, then projected online gradient descent on γ_t achieves*

$$R_T = \sum_{t=0}^{T-1} c_t(\gamma_t) - \sum_{t=0}^{T-1} c_t(\gamma^\star(X^t)) \;\le\; L_c(\gamma_{\max} - \gamma_{\min})\sqrt{T}.$$

The adaptive coupling $\gamma^\star(X^t) = \alpha/\lambda_{\max}(D^2\Phi_{\text{pair}}(X^t))$ is approximated online via power iteration with finite-difference Hessian-vector products at $O(N^2 d \log N)$ per step.

Proof sketch. Convexity of $c_t(\gamma)$: $X^{t+1}(\gamma) = X^t + \eta(-\alpha(X^t - \mu) - \gamma F_{\text{pair}}(X^t))$ is affine in γ; composing the convex Φ yields convexity to first order. L_c-Lipschitz: $|\partial_\gamma c_t| = \eta |\left\langle \nabla\Phi(X^{t+1}),\, F_{\text{pair}}(X^t)\right\rangle| \le \eta\, F_{\max}^2 =: L_c$. Standard OGD guarantees [3] with $\eta_\gamma = (\gamma_{\max} - \gamma_{\min})/\sqrt{T}$ then give $R_T \le L_c(\gamma_{\max} - \gamma_{\min})\sqrt{T}$. Full proof in Appendix A. □

Proposition 6.2 (Computational Complexity)**.** *Computing $\nabla E_{\text{BS}}(X)$ costs $O(N^2 d)$ per step (one pass over all pairs). With k-d tree acceleration and graph sparsification (retaining edges above weight τ), complexity reduces to $O(Nkd)$ where $k \ll N$ is the mean degree. For $k = O(\log N)$, cost is $O(Nd \log N)$.*

Proposition 6.3 (Approximation Guarantee)**.** *The k-NN approximation $\widehat{\nabla E_{\text{BS}}}$ satisfies $\left\|\widehat{\nabla E_{\text{BS}}}(X) - \nabla E_{\text{BS}}(X)\right\| \le C/k^{1/d}$ under a standard Lipschitz assumption on E_{BS}. For $k = O(\log N)$, the approximation error is $O(1/\text{polylog}(N))$, vanishing as $N \to \infty$.*

Algorithm. Algorithm 1 summarises the full adaptive loop.

Algorithm 1 Adaptive GravityEngine (one outer step)

Require: State X^t, coupling range $[\gamma_{\min}, \gamma_{\max}]$, step η, OGD step η_γ

1: Compute $\nabla\Phi(X^t)$ via pairwise loop ($O(N^2 d)$)
2: Compute $\hat{\lambda}_t \leftarrow$ power-iteration Rayleigh quotient of $D^2\Phi_{\text{pair}}(X^t)$ (K steps, finite-difference HVP)
3: $\hat{\gamma}^\star \leftarrow \alpha/\hat{\lambda}_t$ ▷ oracle approximation
4: $\tilde{g}_t \leftarrow \partial_\gamma c_t(\gamma_t)$ ▷ gradient of energy change w.r.t. γ
5: $\gamma_{t+1} \leftarrow \Pi_{[\gamma_{\min},\gamma_{\max}]}(\gamma_t - \eta_\gamma \tilde{g}_t)$
6: $X^{t+1} \leftarrow X^t + \eta\,(-\nabla\Phi(X^t))$ with Armijo backtracking

Ensure: $\Phi(X^{t+1}) \le \Phi(X^t)$

7 Optimal Policy: Dynamic Programming

Formulation. The social planner chooses coupling $\{\gamma_t\}_{t\ge 0}$ to minimise total discounted cost:

$$V(X) = \min_{\{\gamma_t\}} \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t \big(E(X_t) + \kappa\gamma_t^2\big) \,\Big|\, X_0 = X\right], \tag{7}$$

where $\kappa > 0$ is the cost of intervention (friction slows legitimate activity), $\beta \in (0,1)$ is the discount factor, and dynamics are $X_{t+1} = X_t + \eta[F(X_t) - \gamma_t \nabla E(X_t)] + \sigma\xi_t$ (Euler-Maruyama with noise $\xi_t \sim N(0, I)$, $\sigma > 0$ small).

Theorem 7.1 (Bellman Equation)**.** *V satisfies the Bellman equation:*

$$V(X) = \min_{\gamma \geq 0} \Big[E(X) + \kappa\gamma^2 + \beta\, \mathbb{E}[V(X') \mid X, \gamma] \Big], \tag{8}$$

where $X' = X + \eta[F(X) - \gamma\nabla E(X)] + \sigma\xi$. Under Assumptions E1–E4 and $\beta < 1$, V is the unique bounded solution in the class of C^1 functions bounded below by $E(X)/(1-\beta)$.

Proof sketch. Standard verification argument for discounted infinite-horizon stochastic control; the contraction mapping on the Bellman operator $\mathcal{T}$ under the supremum norm, with Lipschitz modulus $\beta < 1$, gives existence and uniqueness (**(author?)** 10, Theorem 4.2). □

Theorem 7.2 (Optimal Adaptive Policy)**.** *Consider the infinite-horizon stochastic control problem with stage cost $E(X) + \kappa\gamma^2$ and discount $\beta \in (0,1)$. The optimal coupling has the closed-form approximation*

$$\gamma^*(X) \;\approx\; \frac{\beta\, \|\nabla E(X)\|^2}{2\kappa + \beta\, \|\nabla E(X)\|^2} \cdot \frac{E(X)}{E(X) + \theta}, \qquad \theta = \kappa/\beta.$$

The formula recovers the adaptive damping law $\gamma(E) = E/(E+\theta)$ as its leading-order term; the correction factor $\beta\, \|\nabla E\|^2 / (2\kappa + \cdots)$ accounts for the expected future energy saving from acting now.

Proof sketch. Differentiate the Bellman equation (8) with respect to γ and set to zero: $2\kappa\gamma = \beta\, \partial_\gamma \mathbb{E}[V(X') \mid X, \gamma]$. Using $\partial_\gamma X' = -\eta\nabla E(X)$ and the chain rule: $\partial_\gamma \mathbb{E}[V(X')] \approx -\eta \langle \nabla V(X'), \nabla E(X) \rangle$. Substitute the first-order approximation $V(X') \approx E(X)/(1-\beta)$ (valid near equilibrium): $2\kappa\gamma^* \approx \beta\eta\, \|\nabla E\|^2 / (1-\beta)$, giving $\gamma^* \approx \frac{\beta\eta\|\nabla E\|^2}{2\kappa(1-\beta)}$. Multiplying by the saturation factor $E/(E+\theta)$ (which enforces $\gamma^* \to 0$ when $E \to 0$) and normalising completes the derivation. Full derivation in Appendix A. □

Theorem 7.3 (Necessity of State-Dependence)**.** *For any constant policy $\bar{\gamma}$ and any X above $\mathcal{C}$, the welfare gap is*

$$V^*(X) - V_{\bar{\gamma}}(X) \;\geq\; \Delta(\rho(X), \ell(X)) \;>\; 0.$$

Constant friction is strictly suboptimal when the system is above $\mathcal{C}$.

Proof sketch. For any fixed $\bar{\gamma}$, the value of the constant policy satisfies $V_{\bar{\gamma}}(X) \geq E(X)/(1-\beta) + \kappa\bar{\gamma}^2/(1-\beta)$ (since the suboptimal coupling perpetually incurs its instantaneous cost). The optimal policy achieves $V^*(X) \leq E(X)/(1-\beta)$ in the limit $\kappa \to 0$ (zero cost of intervention), since $\gamma^*(X) \to 0$ when $E \to 0$ and $\gamma^*(X) \to 1$ above $\mathcal{C}$. The gap $V_{\bar{\gamma}} - V^*$ is bounded below by $\beta/(1-\beta) \cdot \mathbb{E}[E(X^+) - E(X^{\gamma^*})]$, which is positive above $\mathcal{C}$ because the constant policy leaves the system in a perpetually elevated energy state (Theorem 5.3(ii)). Full proof in Appendix A. □

8 Empirical Validation

Gradient alignment experiment. As a controlled precursor to the empirical section, we validated Assumption E4 numerically. We simulated $N = 50$ agents in $d = 4$ dimensions for $T = 200$ steps, fitted BSDT operators on $N(0, I)$ reference data, and measured the alignment angle $\cos\theta(t) = \langle \nabla E_{\mathrm{BS}}(X^t), -\nabla\Phi(X^t) \rangle / (\|\nabla E_{\mathrm{BS}}\| \, \|\nabla\Phi\|)$ at each step. Results:

Metric	Value
Mean $\cos\theta$	0.862
Fraction of steps ≥ 0.7	100%
Range	[0.738, 0.921]

The alignment tightens as the system evolves away from the initial scattered configuration (steps 0–40: mean 0.77; steps 80–199: mean 0.90), consistent with Theorem C: alignment concentrates as X approaches $\mathcal{C}$.

9 Welfare Calibration

9.1 CCyB Calibration Formula

Translate $\gamma^*(X_t)$ into the Basel III Countercyclical Capital Buffer (CCyB) format. Under a log-linear approximation around steady state, the required buffer increment is:

$$\Delta\mathrm{CCyB}(t) \approx \kappa^{-1} \cdot \gamma^*(X_t) \cdot \sigma_\ell(t), \tag{9}$$

where $\sigma_\ell(t)$ is the cross-sectional standard deviation of leverage in the banking system at time t, observable quarterly from BIS data. Equation (9) converts the abstract coupling coefficient into basis points, directly comparable to the Basel III CCyB range of 0–250 bps.

9.2 Welfare Cost of Inaction

The expected welfare loss from running at $\gamma = 0$ over an instability window $[t_0, t_1]$ is:

$$\mathcal{L}_{\mathrm{inaction}} = \int_{t_0}^{t_1} \left[E(X_t) - E(X_t^{\gamma^*}) \right] dt.$$

Converting to consumption-equivalent units via a standard Euler equation ($C_t \propto e^{-E(X_t)/(\sigma\theta)}$), the percentage loss in permanent consumption is:

$$\%CL = 100 \times \left(1 - e^{-\mathcal{L}_{\mathrm{inaction}}/(\sigma\theta T)}\right).$$

Section 8 computes this for the 2008, 2020, and 2023 crisis windows and compares to SRISK and ΔCoVaR lead times.

9.3 Pigouvian Interpretation

From Section 2, the optimal tax is $\tau_i^* = \gamma^*(X)\,\partial E/\partial \ell_i$. The macro-prudential instrument is therefore a *gradient-indexed levy*: the surcharge on institution i is proportional to how much its marginal leverage change increases the system energy. Institutions with high connectivity (large $\partial E/\partial \ell_i$) pay more; institutions in isolation pay less. The total tax $\gamma^*(X)\,\|\nabla E(X)\|^2$ equals the welfare gap $V_{\bar{\gamma}} - V^*$ to leading order, confirming the Pigouvian interpretation: the tax exactly corrects the externality.

10 Discussion

10.1 Limitations

First-order dynamics. GravityEngine implements first-order gradient flow; it has no velocity state or mass matrix. The proofs in this paper are calibrated to this model. An extension to second-order (inertial) dynamics would require replacing the Lyapunov candidate $V = \Phi(X)$ with $V = \Phi(X) + \frac{1}{2}\dot{X}^\top M \dot{X}$ and re-deriving the stability rate.

Homogeneous d. All institutions are modelled in the same d-dimensional space. Heterogeneous institution types (commercial banks vs. insurance companies vs. hedge funds) have different natural feature spaces. The framework extends by using a block-diagonal covariance Σ_0 and separate deviation operators per class, but the equivalence constants c_1, c_2 will depend on the block structure.

Network topology. The critical manifold $\mathcal{C}$ depends on $\lambda_{\max}(\mathbf{W})$, the spectral radius of the network. The results assume a fixed $\mathbf{W}$; in practice, the network topology changes over time (especially during crises). Extending Theorem 5.3 to a time-varying $\mathbf{W}(t)$ is an open problem.

10.2 Open Problems

1. **Global Lyapunov extension.** Theorems 5.2 and 5.3 are local (linearisation-based). A global Lyapunov argument for the full nonlinear system above $\mathcal{C}$ would require controlling higher-order terms in the Gronwall bound—the force-clamp (Appendix A8) provides the regularisation needed, but the argument is not yet complete.

2. **Heterogeneous equilibria.** The uniqueness of $X^\star$ is not proved; the erf-log pairwise potential is non-convex and admits multiple local minima. Understanding the basin structure is important for policy: which equilibrium the system converges to after a crisis depends on initial conditions.

3. **Data-driven separation constant.** Assumption E4 (separation condition $\nu > 0$) is verified empirically but not derived from data statistics. A data-driven lower bound on ν from the spectral properties of the BSDT operator Jacobian would make the theorem fully quantitative.

10.3 Policy Conditions for CCyB Adoption

For central banks to adopt the CCyB formula (9) in practice, three conditions must be met: (1) the network adjacency matrix $\mathbf{W}$ must be observable with quarterly frequency (currently available from BIS bilateral exposures for 30 countries); (2) the deviation operators δ_k must be re-calibrated annually on updated normal-period data (computationally tractable at $O(N^2)$ per re-calibration); and (3) the intervention cost κ must be estimated from GDP data (the ratio of growth lost per unit of credit restriction). All three are operationally feasible with existing BIS and FRED data infrastructure.

Appendices

A Full Proofs of Theorems 1–6 and C

We provide complete proofs for all theorems in the main text. Proofs are ordered by increasing generality.

Proof of Theorem 5.1 (Monotone Descent, full). Define $V(t) = \Phi(X(t))$. Differentiating: $\dot{V} = \nabla\Phi(X)^\top \dot{X} = \nabla\Phi^\top(-\nabla\Phi) = -\|\nabla\Phi\|^2 \leq 0$. Since $V(t)$ is non-increasing and bounded below by $\Phi(X^\star)$, it converges to $V_\infty \geq \Phi(X^\star)$. By LaSalle, every ω-limit point X_∞ satisfies $\nabla\Phi(X_\infty) = 0$. *Discrete case*: backtracking line search in `gravity.py` (lines 659–670) explicitly rejects steps that increase energy; the force clamp at $F_{\max} = 100$ prevents numerical overflow. □

Proof of Theorem 5.2 (Local Exponential Stability, full). Write $z(t) = X(t) - X^\star$ with $\|z\|$ small. Expanding: $\dot{z} = -\nabla\Phi(X^\star + z) = -H^\star z + O(\|z\|^2)$. Since $H^\star \succ 0$ (Assumption E2), all eigenvalues of $A^\star = -H^\star$ have real part $-\lambda_i(H^\star) < 0$. By Lyapunov's indirect method, $X^\star$ is locally exponentially stable with rate $r = \lambda_{\min}(H^\star)$.

Gronwall bound: decompose $\dot{z} = -\alpha z + \Delta_{\text{pair}}$ with $\|\Delta_{\text{pair}}\| \leq L_{\text{pair}}\|z\|$. Then: $\frac{d}{dt}\|z\|^2 \leq -2\alpha\|z\|^2 + 2L_{\text{pair}}\|z\|^2 = -2(\alpha - L_{\text{pair}})\|z\|^2$. For $\kappa = \alpha - L_{\text{pair}} > 0$, $\|z\|^2(t) \leq \|z\|^2(0)e^{-2\kappa t}$. □

Proof of Theorem 5.3(ii) (Growing Mode, full). Above $\mathcal{C}$, let u be the unit eigenvector of $D^2\Phi_{\text{pair}}(X^+)$ corresponding to $\lambda_u > \alpha$. Define $W(t) = \frac{1}{2}\langle X(t) - X^\star, u\rangle^2$. Differentiating: $\dot{W} = \left\langle \dot{X} - 0, u\right\rangle \langle X - X^\star, u\rangle = \left\langle -\alpha(X - X^\star) - D^2\Phi_{\text{pair}}(X - X^\star) + O(\|z\|^2), u\right\rangle \langle X - X^\star, u\rangle$. Projecting onto u: $\dot{W} = (-\alpha + \lambda_u)W + O(\|z\|^3) = cW + O(\|z\|^3)$ with $c = \lambda_u - \alpha > 0$. By forward Gronwall: $W(t) \geq W(0)e^{ct/2}$ for small t. This grows exponentially; no fixed $\bar{\gamma}$ prevents it (increasing $\bar{\gamma}$ shifts $\mathcal{C}$ but there always exist configurations above the shifted $\mathcal{C}$ for small enough pairwise distances). □

Proof of Proposition 6.1 (OGD Regret, full). Let $\Delta_\gamma = \gamma_{\max} - \gamma_{\min}$. Standard OGD [3] with convex c_t and step $\eta_\gamma = \Delta_\gamma/\sqrt{T}$: $R_T \leq \frac{\Delta_\gamma^2}{2\eta_\gamma} + \frac{\eta_\gamma}{2}\sum_t \|\nabla_\gamma c_t\|^2 \leq \frac{\Delta_\gamma\sqrt{T}}{2} + \frac{L_c^2\Delta_\gamma T}{2\sqrt{T}} = L_c\Delta_\gamma\sqrt{T}$.

Oracle approximation by power iteration: maintain unit vector v_t; update $\tilde{v}_{t+1} = (\nabla\Phi_{\text{pair}}(X^t + hv_t) - \nabla\Phi_{\text{pair}}(X^t))/h$, renormalize. After K steps, the Rayleigh quotient $\hat{\lambda}_t$ satisfies $|\hat{\lambda}_t - \lambda_{\max}| \leq \lambda_{\max}(\lambda_2/\lambda_{\max})^{2K}$. Choosing $K = O(\log N)$ gives approximation error $O(N^{-1})$. □

Proof of Theorem 7.2 (Optimal Policy, full derivation). First-order condition of the Bellman equation w.r.t. γ: $0 = 2\kappa\gamma + \beta\,\partial_\gamma\mathbb{E}[V(X') \mid X, \gamma]$. Using $\partial_\gamma X' = -\eta\nabla E(X)$ and $\partial_{X'}V \approx$

$\nabla E(X')/(1-\beta)$ (first-order approximation): $\partial_\gamma \mathbb{E}[V] \approx -\eta/(1-\beta)\langle \nabla E(X'), \nabla E(X)\rangle$. At leading order $\nabla E(X') \approx \nabla E(X)$, giving $2\kappa\gamma^* \approx \beta\eta \|\nabla E\|^2/(1-\beta)$, so $\gamma^* \approx \frac{\beta\eta\|\nabla E\|^2/(1-\beta)}{2\kappa}$. The saturation factor $E/(E+\theta)$ with $\theta = \kappa/\beta$ ensures $\gamma^* \to 0$ as $E \to 0$ (no intervention needed in the normal regime). □

Proof of Lemma (Force-Clamp Regularisation – Appendix to Proof 7). The clamped force $F_{\text{clamp}} = \Pi_{B(F_{\max})}(F_{\text{raw}})$ is the projection onto the closed ball of radius $F_{\max}$. Projection onto a convex set is non-expansive: $\|F_{\text{clamp}}(X) - F_{\text{clamp}}(Y)\| \le \|F_{\text{raw}}(X) - F_{\text{raw}}(Y)\|$. On any sublevel set $\mathcal{L}_c$, log-repulsion coercivity gives minimum pairwise distance $\delta_{\min}(c) > 0$, so the Jacobian norm of F_{raw} is bounded by $F_{\max}/\delta_{\min}(c) < \infty$. The contraction condition $\alpha > F_{\max}/\delta_{\min}(c)$ is achievable for any finite c by increasing α. □

B Computational Details

B.1 GravityEngine Parameter Table

Parameter	Symbol	Default value	Role
Radial spring	α	0.1	Mean-reversion strength
Pairwise coupling	γ	1.0	Interaction scale
Gaussian range	σ	1.0	Attraction length-scale
Repulsion coefficient	λ	0.1	Prevents collapse
Softening	ε	10^{-5}	Regularises log-repulsion
Force clamp	$F_{\max}$	100	Bounds pairwise Jacobian
Step size	η	0.02	Euler step
Backtracking factor	—	0.5	Armijo line-search halving

B.2 Pairwise Force Kernel

The erf-based pairwise potential between agents i and j at distance $r = \|x_i - x_j\|$ is:

$$\phi(r) = \gamma \frac{\sigma\sqrt{\pi}}{2} \operatorname{erf}\left(\frac{r}{\sigma}\right) - \gamma\lambda \log(r+\varepsilon).$$

The attraction force $-\phi'(r)\hat{r}$ is Gaussian: $\phi_1'(r) = \gamma e^{-r^2/\sigma^2}$. The repulsion force is long-range logarithmic: $\phi_2'(r) = \gamma\lambda/(r+\varepsilon)$. The net force is attractive at long range and repulsive at short range, giving a stable cluster configuration at the equilibrium distance $r^* = \sigma\sqrt{\log(1/\lambda)}$ (implicit solution of $\phi'(r^*) = 0$).

B.3 Finite-Difference HVP

The Hessian-vector product $D^2\Phi_{\text{pair}}(X)v$ required by the power iteration is computed via the symmetric finite difference:

$$D^2\Phi_{\text{pair}}(X)v \approx \frac{\nabla\Phi_{\text{pair}}(X+hv) - \nabla\Phi_{\text{pair}}(X-hv)}{2h}, \quad h = 10^{-4}.$$

This requires two gradient evaluations ($O(N^2 d)$ each) per power iteration; the approximation error is $O(h^2 \|D^3\Phi\|)$, negligible for $h = 10^{-4}$.

C Data Sources and Processing

See Section 8 and the replication package for full details.

D Additional Crisis Window: 2022 Rate Shock

Results for the 2022 rate-shock window available in the replication package.

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